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Single Piece, Open Skirt Piston Reuse Guidelines

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This Service Bulletin applies inspection criteria for single piece, open skirt pistons in Signature™, ISX, QSX15, ISX11.9 CM2250, ISX15 CM2250 ECF, QSX11.9 CM2250 ECF, PowerGen QSX15 CM2250, PowerGen QSX15 CM2250 ECF, QSX15 CM2350 X105, and QSX15 CM2350 X106 engines. Cummins Inc. distributors and dealers are encouraged to consult these guidelines when seeking to determine if used single piece, open skirt pistons are acceptable for reuse.

For ISX and QSX15 engines, see the following procedure in Service Manual, Signature™, ISX, and QSX15, Bulletin 3666239, for reuse guidelines for bowl cracks, skirt pin bore, and the ring grooves. Refer to Procedure 001-043 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-043.html>)

Use the following procedure for reuse guidelines for bowl cracks, skirt pin bore, and ring grooves for ISX12/ISX11.9 CM22250 engines in the ISX12/ISX11.9 CM2250 Service Manual, Bulletin 2883445. Refer to Procedure 001-043 in Section 1. (</qs3/pubsys2/xml/en/procedures/133/133-001-043-tr-mixab.html>)

Use the following procedure for reuse guidelines for bowl cracks, skirt pin bore, and ring grooves for ISX15 CM22250 engines in the ISX15 CM2250 Service Manual, Bulletin 4022250. Refer to Procedure 001-043 in Section 1. (</qs3/pubsys2/xml/en/procedures/132/132-001-043-tr-mixar5.html>)

Use the following procedure for reuse guidelines for bowl cracks, skirt pin bore, and ring grooves for QSX15 CM2250 ECF engines in the QSX15 CM2250 ECF Service Manual, Bulletin 2883557. Refer to Procedure 001-043 in Section 1. (</qs3/pubsys2/xml/en/procedures/132/132-001-043-tr-vqxi6.html>)

Use the following procedure for reuse guidelines for bowl cracks, skirt pin bore, and ring grooves for QSX11.9 CM2250 ECF engines in the QSX11.9 CM2250 ECF Service Manual, Bulletin 2883561. Refer to Procedure 001-043 in Section 1. (</qs3/pubsys2/xml/en/procedures/133/133-001-043-tr-vqxb.html>)

Use the following procedure for reuse guidelines for bowl cracks, skirt pin bore, and ring grooves for PowerGen QSX15 CM22250 engines in the PowerGen QSX15 CM2250 ECF Service Manual, Bulletin 4310661. Refer to Procedure 001-043 in Section 1. (</qs3/pubsys2/xml/en/procedures/132/132-001-043-tr-vqyg5a.html>)

Use the following procedure for reuse guidelines for bowl cracks, skirt pin bore, and ring grooves for PowerGen QSX15 CM22250 engines in the PowerGen QSX15 CM2250 Service Manual, Bulletin 4310664. Refer to Procedure 001-043 in Section 1. (</qs3/pubsys2/xml/en/procedures/132/132-001-043-tr-vqyg5na.html>)

Use the following procedure for reuse guidelines for bowl cracks, skirt pin bore, and ring grooves for QSX15 CM2350 X105 engines in the QSX15 CM2350 X105 Service Manual, Bulletin 4332667. Refer to Procedure 001-043 in Section 1 (</qs3/pubsys2/xml/en/procedures/250/250-001-043.html>).

Use the following procedure for reuse guidelines for bowl cracks, skirt pin bore, and ring grooves for QSX15 CM2350 X106 engines in the QSX15 CM2350 X106 Service Manual, Bulletin 4332666. Refer to Procedure 001-043 in Section 1 (</qs3/pubsys2/xml/en/procedures/221/221-001-043.html>).

The photographs below illustrate the reuse guidelines for second land wear. The first two illustrations display second land polish and are OK for reuse.



Second land polish

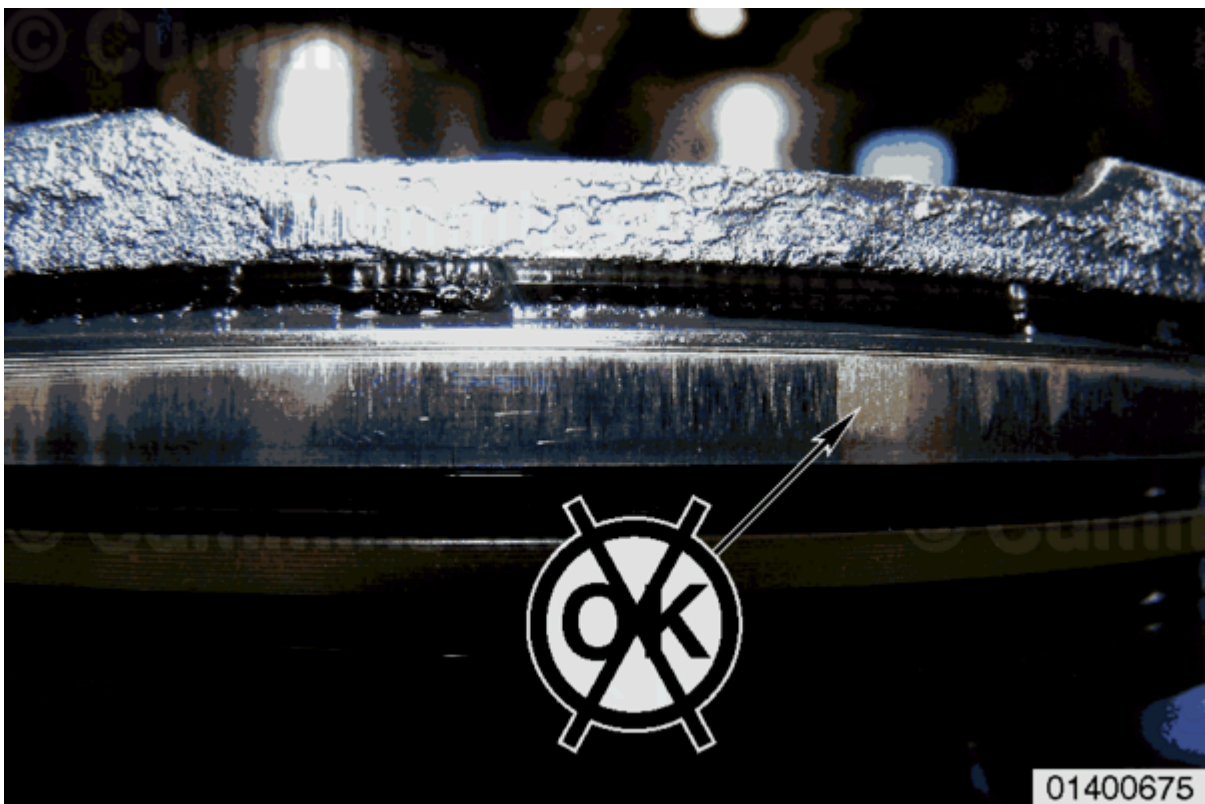


Second land polish

The following three photographs illustrate pistons that are **not** acceptable for reuse. These pistons have second land scuffing. This is visible by the presence of vertical scratches.



Light scuffing of second land



Moderate scuffing of second land



Heavy scuffing of second land

The following photographs illustrate the reuse guidelines for piston skirt wear. The first two photographs display light piston skirt scratching and are OK for reuse.



Light skirt scratching



Light skirt scratching

The following three photographs below illustrate piston skirts which are **not** acceptable for reuse. These pistons have skirt scuffing that is apparent by the presence of heavy vertical scratching. In general, if these scratches can be felt with a fingernail, the piston should be replaced.



Heavy scratching of piston skirt



Heavy scratching of piston skirt



Heavy scuffing of piston skirt

Document History

Date	Details
xxxx-xx-xx	Module Created
2012-4-13	Modified to include stationary regen differences specific to Industrial
2014-9-29	Add QSX15 CM2350

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